

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2025-26 (NEP)

COMPUTER SCIENCE**Paper Code: CSCMJ44 : Computer Organization and Architecture**

Time: 2:30 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions in all including Question No. 1, which is compulsory. Terms and abbreviations have their standard meaning. Figures on the right-hand side margin indicate Max. Mark for each question.*

Question 1. (Compulsory)

- (a) A 32-bit physical address is divided into three fields: Tag, Set Index, and Block Offset of 21 bits, 7 bits, and 4 bits, respectively, for use in a 2-way set-associative cache. The cache is initially empty. The following byte addresses (in decimal) are accessed in the given order: 0, 8, 32, 144, 272, 208, 2048, 60, 152, 4120, 224, 3100. The cache uses LRU (Least Recently Used) replacement policy. Determine the final state of each valid entry in the cache by listing the pairs of (Set Index, Tag) in decimal. [6]
- (b) Explain the difference between hardwired control unit and microprogrammed control unit. [4]
- (c) Compare RISC and CISC architectures. [4]

Question 2.

- (a) Design a 4-bit combinational decrement circuit using four full adder circuits. [4]
- (b) What is an instruction cycle, and write the phases of the Instruction cycle? [3]
- (c) A computer has 16 registers and an ALU capable of performing 32 operations. All the registers and the ALU are connected through a common bus system. [7]
 - (i) Formulate the control word required to specify a microoperation.
 - (ii) Specify the number of bits required for each field of the control word, and provide a general encoding scheme for the fields.
 - (iii) Determine the bit pattern of the control word required to perform the following microoperation: $R4 \leftarrow R5 + R6$.

Question 3.

- (a) Explain the role of status register in CPU with suitable examples. [7]

- (b) The stack is a memory stack that grows downward (i.e., the Stack Pointer (SP) is decremented before a value is pushed onto the stack and incremented after a value is popped from the stack). The top of the stack contains the value 5320, and the Stack Pointer (SP) contains 3560. A two-word CALL subroutine instruction is stored in memory, with the opcode at address 1120 and its address field (6720) at address 1121. Determine the contents of the Program Counter (PC), Stack Pointer (SP), and the top of the stack at each of the following stages: [7]
- (i) Before the CALL instruction is fetched from memory.
 - (ii) After the CALL instruction has been executed.
 - (iii) After the subroutine returns (i.e., after execution of the RETURN instruction).
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Question 4.

- (a) You are given 64×8 RAM chips and 128×8 ROM chips. Using these, construct a physical memory of size 256 bytes and draw a block diagram showing how this memory is connected to the CPU. [7]
- (b) Explain the concept of memory hierarchy in a computer system. Why is a memory hierarchy required? [4]
- (c) An address space is specified by 24 bits and the corresponding memory space by 16 bits. How many words are there in the address space and memory space? If a page consists of 2k words, how many pages and blocks are there in the system? [3]
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Question 5.

- (a) What is Daisy-chain priority. Explain it with suitable block diagram. [6]
- (b) What is the difference between vectored interrupt and non-vectored interrupt? [4]
- (c) What is the difference between isolated I/O and memory-mapped I/O? [4]
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Question 6.

- (a) Explain the working of DMA (Direct Memory Access) with the block diagram. [7]
- (b) What are the basic differences among a branch instruction, a call sub-routine instruction and a program interrupt? [7]
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Question 7. Write short notes on:

[14]

- (a) Stored program concept,
 - (b) Zero-address instruction,
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- (c) Bootstrap Loader,
- (d) Multi-processor system, and
- (e) Input-output interface